CLW-Fisher: Prior-Guided Test-Time Model Merging with Co-acting Layer-Wise Fisher Adaptation for Open-World Streams

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Abstract

Test-Time Model Merging (TTMM) has emerged as a parameter-efficient paradigm for dynamically combining specialized expert models in weight space on-the-fly to handle non-stationary, unlabeled test streams. However, existing open-world TTMM frameworks suffer from fundamental bottlenecks: they rely on feature prototype routing with a fixed softmax temperature, which leads to over-smoothed or highly erroneous routing coefficients under environmental noise, and they reset merging parameters to 0.5 at each batch, ignoring the routing prior and slowing optimization convergence. Furthermore, standard layer-wise adaptation allows layers to adapt completely independently, which can cause representational misalignment and catastrophic activation mismatch across the network. To address these limitations, we propose a unified framework called CLW-Fisher (Co-acting Layer-Wise Fisher adaptation) integrated with Prior-Guided Initialization (PG-Init) and a novel Self-Calibrated Temperature Scaling (SCTS) mechanism. SCTS is a mathematically principled, parameter-free temperature formulation that dynamically calibrates the routing temperature based on the absolute prototype distance gap, achieving scale invariance and a pre-defined maximum routing confidence. Simultaneously, PG-Init translates this dynamic prior directly into the initial weight parameters, accelerating convergence. Finally, CLW-Fisher models merging parameters as a global consensus logit with layer-wise offsets preconditioned by Joint Fisher Information, stabilized with a consensus coherence penalty. Our extensive evaluations on non-stationary vision streams show that CLW-Fisher with SCTS achieves outstanding performance, outperforming fixed-temperature resetting baselines by up to +19.37% absolute classification accuracy under noise while successfully preventing representational feedback loops.

1. Introduction

The rapid progress of deep learning has led to a paradigm shift towards training massive foundation models and fine-tuning them on specific target domains, resulting in a vast collection of specialized expert networks (Wortsman et al., 2022; Radford et al., 2021). Integrating these expert networks into a single, cohesive, multi-task architecture without incurring prohibitive retraining costs has motivated the development of model merging and weight-space averaging techniques (Ilharco et al., 2023; Matena & Raffel, 2022; Ainsworth et al., 2023). By interpolating parameters directly, model merging bypasses the need for joint training data, maintaining parameter efficiency while achieving remarkable multi-task capabilities.

Recently, this concept has been extended to the streaming inference phase via Test-Time Model Merging (TTMM) (Yang et al., 2024; Hubotter et al., 2025). In practical applications, deep networks encounter non-stationary target streams where the active task distribution shifts continuously over time. TTMM dynamically interpolates the weight parameters of specialized expert networks on-the-fly to match the active task distribution of an unlabeled, non-stationary test stream:

$$\bar{\theta}_t = \lambda^{(t)}\theta_1 + (1 - \lambda^{(t)})\theta_2 \tag{1}$$

where θ_1 and θ_2 represent the weights of two expert networks trained on separate tasks, and $\lambda^{(t)} \in [0, 1]$ represents the dynamic merging coefficient at time step t.

Despite initial successes, adapting merging coefficients under realistic, open-world deployment remains a

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

fundamental challenge. State-of-the-art open-world TTMM frameworks utilize feature prototype routing to detect novel domains and compute merging coefficients (Anonymous, 2026c;b). However, these approaches employ a fixed temperature parameter in their routing softmax, which fails to account for the dynamic levels of uncertainty and noise present in real-world test streams.

Specifically, when a known task undergoes environmental noise (e.g., Gaussian noise), the distance from the incoming test features to all pre-computed prototypes increases proportionally (Hendrycks & Dietterich, 2019). This pushes the absolute similarity values closer together, causing a fixed-temperature routing softmax to output a near-uniform coefficient distribution (e.g., $[0.5, 0.5]$). This premature blending of experts causes severe activation mismatches and representational decay on known tasks. Conversely, when a novel task arrives, the model's coefficients are highly vulnerable to the "feedback loop trap" (Anonymous, 2026c), where unconstrained optimization collapses the coefficients toward an arbitrary, overconfident out-of-distribution expert, destroying model utility.

Furthermore, we identify another critical and previously unaddressed bottleneck in existing literature: standard test-time adaptation pipelines initialize the merging parameter logits to 0.0 (corresponding to a flat 0.5/0.5 mixture) at every incoming batch. Starting the optimization from the midpoint on each batch ignores the strong routing evidence already available, leading to sluggish convergence, sub-optimal adaptation within limited gradient steps, and susceptibility to representational collapse.

Importantly, when executing layer-wise adaptation to address the highly heterogeneous sensitivities across network layers, existing approaches allow each layer's merging parameters to adapt completely independently. This unconstrained layer-wise optimization leads to a severe issue of "representational misalignment," where adjacent layers are merged with highly divergent coefficients (e.g., an early convolutional layer is merged with $\lambda = 0.9$ while the subsequent layer is merged with $\lambda = 0.1$). This mismatch disrupts the continuous representational flow through the network, causing activation mismatches that severely degrade model performance.

To resolve these fundamental bottlenecks, we propose CLW-Fisher (Co-acting Layer-Wise Fisher-preconditioned adaptation) integrated with Self-Calibrated Temperature Scaling (SCTS) and Prior-Guided Initialization (PG-Init). Our framework introduces three key contributions:

- Self-Calibrated Temperature Scaling (SCTS): We formulate a mathematically principled, parameter-free temperature formulation. Rather than relying on heuristic base temperatures or scaling powers, SCTS dynamically calibrates the routing temperature based on the absolute distance difference (gap) between the closest and second-closest expert prototypes. SCTS achieves perfect scale invariance and allows a pre-defined maximum routing confidence on known tasks, while naturally uniformizing routing priors on novel domains.
- Prior-Guided Parameter Initialization (PG-Init): We propose initializing the learnable merging parameter logits based on the computed routing prior using the inverse sigmoid (logit) function. This aligns the initial optimization state with the routing prior, accelerating convergence and substantially boosting adaptation accuracy.
- Co-acting Layer-Wise Fisher Adaptation (CLW-Fisher): Rather than allowing independent layer-wise updates, we formulate a co-acting layer-wise optimization framework. We model each layer's merging coefficient as a combination of a global consensus logit and a layer-specific offset, regularized by a consensus coherence penalty. We scale each offset's update rate inversely to its precomputed Joint Fisher Information sensitivity, shielding highly sensitive layers from representation drift while maintaining a strong, unified representational consensus.

We evaluate CLW-Fisher on a challenging non-stationary vision stream benchmark containing MNIST (LeCun et al., 1998), FashionMNIST (Xiao et al., 2017), and KMNIST (Clanuwat et al., 2018) (treated as a novel domain) under both clean and corrupted environments. Our empirical results demonstrate that our unified framework achieves outstanding performance, consistently outperforming standard fixed-temperature resetting baselines by up to +19.37% absolute classification accuracy on corrupted streams while maintaining robust stability across all environments.

2. Related Work

Model Merging: Model merging refers to the weight-space combination of multiple fine-tuned neural networks to produce a single multi-task model without expensive joint retraining. Foundational works like

Model Soups (Wortsman et al., 2022) showed that averaging fine-tuned models starting from a shared pre-trained initialization consistently improves robustness and accuracy. Task Arithmetic (Ilharco et al., 2023) demonstrated that task-specific vectors can be added or subtracted to edit model behaviors. Fisher Weighted Merging (Matena & Raffel, 2022) and Git Re-Basin (Ainsworth et al., 2023) further advanced this by aligning weights and scaling updates based on parameter sensitivities. These methods operate primarily offline, requiring a pre-defined mixture of experts before deployment.

Test-Time Adaptation (TTA): TTA seeks to adapt a pre-trained model to target domain shifts during inference using only unlabeled test data. Prevalent TTA methods such as TENT (Wang et al., 2021) minimize prediction entropy, while EATA (Niu et al., 2022), CoTTA (Wang et al., 2022), and SAR (Niu et al., 2023) introduce regularization, stable prediction anchoring, and feedback loop prevention to mitigate catastrophic forgetting and representation collapse under continuous, non-stationary shifts. Other methods, such as GDA (Tsai et al., 2024) and MEMO (Yuan et al., 2023), focus on parameter-efficient updates of key projection matrices.

Test-Time Model Merging (TTMM): Bringing model merging to the streaming setting, TTMM dynamically updates weight-space merging coefficients on-the-fly to adapt to incoming test distributions (Yang et al., 2024; Hubotter et al., 2025). Recent works have identified crucial bottlenecks in TTMM, including the omission of Batch Normalization running statistics (Anonymous, 2026a). To operate in open-world scenarios where novel tasks arrive dynamically, frameworks like IGGS-OW (Anonymous, 2026c) and FP-OW (Anonymous, 2026b) precompute prototypes and utilize diagonal Fisher Information to precondition coefficient updates. However, these methods rely on a fixed routing temperature and reset adaptation parameters on each batch, leaving them vulnerable to noise, slow convergence, and overconfidence. Our proposed CLW-Fisher directly bridges this gap.

3. Methodology

We consider the open-world TTMM setting where a stream of unlabeled test batches $X^{(1)}, X^{(2)}, \ldots, X^{(T)}$ arrives sequentially. We assume access to a set of specialized expert models $\mathcal{M} = \{M_k\}_{k=1}^{K}$ fine-tuned from a shared base initialization M_{base}. Our goal is to merge these experts into a single unified network $\bar{M}_t$ characterized by merging coefficients $\lambda^{(t)}$ on-the-fly, maximizing classification accuracy while avoiding representational collapse.

3.1. Prototype-Based Expert Routing

Following standard open-world model merging literature (Anonymous, 2026c), we precompute class-wise feature prototypes $\mathcal{P}_{k,c}$ for each known expert $k \in \{1, \ldots, K\}$ and class $c \in \{1, \ldots, C\}$ on a small offline calibration set $D_{cal,k}$. Let $\Phi_k(\cdot)$ denote the frozen feature extraction layer of expert k. The prototype for class c in expert k is defined as:

$$\mathcal{P}_{k,c} = \frac{1}{|D^c_{cal,k}|} \sum_{x \in D^c_{cal,k}} \Phi_k(x) \tag{2}$$

When an unlabeled test batch $X^{(t)} = \{x_1, \ldots, x_B\}$ of size B arrives at step t, we project its features into the Static Unified Space of each expert k and compute the average minimum Euclidean distance to the precomputed prototypes:

$$\bar{D}_k(X^{(t)}) = \frac{1}{B} \sum_{i=1}^{B} \min_{c \in \{1,\ldots,C\}} \|\Phi_k(x_i) - \mathcal{P}_{k,c}\|_2^2 \tag{3}$$

The prototype-based similarity $S_k(X^{(t)})$ to expert k is characterized as the negative average distance, $S_k(X^{(t)}) = -\bar{D}_k(X^{(t)})$. The routing probability $w_k(X^{(t)})$ for expert k is computed using a softmax distribution:

$$w_k(X^{(t)}) = \frac{\exp(S_k(X^{(t)})/\tau)}{\sum_{j=1}^{K} \exp(S_j(X^{(t)})/\tau)} \tag{4}$$

where τ is the routing softmax temperature.

3.2. Self-Calibrated Temperature Scaling (SCTS)

While Ratio-Based Dynamic Temperature Scaling (R-DTS) successfully adjusts routing softmax temperatures under noise, it still relies on heuristic parameters like the base temperature τ_{base}, scaling sensitivity power α, and clamping bounds $[\tau_{min}, \tau_{base}]$. Choosing these parameters across different test domains can be highly challenging in open-world settings.

To overcome these limitations and provide a mathematically principled, parameter-free formulation, we propose Self-Calibrated Temperature Scaling (SCTS). SCTS computes the routing softmax temperature dynamically and adaptively on-the-fly directly from the absolute distance difference (gap) between the closest and second-closest expert prototypes.

Let $\bar{D}_{min}(X^{(t)})$ and $\bar{D}_{second}(X^{(t)})$ represent the minimum and second-minimum average prototype distances across all expert networks for the incoming

test batch $X^{(t)}$, and let $\Delta(X^{(t)}) = \bar{D}_{second}(X^{(t)}) - \bar{D}_{min}(X^{(t)})$ represent the absolute prototype distance gap. We define the self-calibrating temperature as:

$$\tau_{self}(X^{(t)}) = \frac{\Delta(X^{(t)})}{s} + \epsilon_{stab} \tag{5}$$

where $s > 0$ represents a target confidence scale factor, and $\epsilon_{stab} > 0$ is a small numerical stability constant.

Mathematical Invariance and Scale Calibration: SCTS exhibits remarkably elegant mathematical properties under softmax routing. Substituting $\tau_{self}(X^{(t)})$ into the routing softmax (Equation (4)) for a two-expert setting with absolute distance gap $\Delta = \Delta(X^{(t)})$ yields the routing prior $w_{min}(X^{(t)})$ for the closest expert:

$$\begin{aligned} w_{min}(X^{(t)}) &= \frac{e^{-\bar{D}_{min}/\tau_{self}}}{e^{-\bar{D}_{min}/\tau_{self}} + e^{-\bar{D}_{second}/\tau_{self}}} \\ &= \frac{1}{1 + \exp\left(-\Delta/\tau_{self}\right)} \\ &= \frac{1}{1 + \exp\left(-\frac{\Delta}{\Delta/s + \epsilon_{stab}}\right)} \end{aligned} \tag{6}$$

Analyzing the asymptotic behavior of SCTS under different streaming environments reveals its exceptional properties:

1. Known Tasks with High Confidence ($\Delta \gg \epsilon_{stab}$): The routing prior simplifies directly to:

$$w_{min}(X^{(t)}) \approx \frac{1}{1 + e^{-s}} \tag{7}$$

   Thus, by selecting the scale factor s, we can perfectly and explicitly pre-determine the routing confidence of our system. For example, selecting $s = 3.0$ yields a maximum routing confidence of exactly $\sigma(3.0) \approx 95.26\%$ for the correct expert, completely independent of the absolute magnitude or scale of the features!

2. Novel/Unknown Tasks ($\Delta \to 0$): As the gap shrinks, the term inside the exponent approaches zero:

$$w_{min}(X^{(t)}) \to \frac{1}{1 + e^{0}} = 0.5 \tag{8}$$

   This automatically uniformizes the routing prior under high uncertainty, preventing the overconfident updates and feedback loops that degrade other methods.

3. Environmental Noise (Intermediate Gap): SCTS dynamically interpolates the softmax temperature based on the ratio of the batch-wise distance gap to the stability constant ϵ_{stab}, smoothly adjusting confidence to prevent decision-boundary decay.

3.3. Temporal-Smoothed Self-Calibrated Temperature Scaling (TS-SCTS)

While SCTS delivers a mathematically principled and parameter-free temperature formulation, its computation relies entirely on the absolute prototype distance gap $\Delta(X^{(t)})$ of the current incoming batch. In real-world streaming environments under extremely small batch sizes (e.g., $B = 16$) or extreme corruptions, the batch-wise distance gap exhibits high statistical variance. This batch-wise noise can destabilize the temperature estimation and propagate to the routing prior.

To filter out high-frequency noise and exploit the temporal coherence of sequential streaming data (where adjacent batches often belong to the same task or domain), we propose Temporal-Smoothed Self-Calibrated Temperature Scaling (TS-SCTS). We maintain a running Exponential Moving Average (EMA) of the absolute distance gap across streaming steps:

$$\bar{\Delta}^{(t)} = \gamma \bar{\Delta}^{(t-1)} + (1 - \gamma)\Delta(X^{(t)}) \tag{9}$$

where $\gamma \in [0, 1)$ is the smoothing coefficient (we set $\gamma = 0.8$ in our experiments). The smoothed temperature is then computed as:

$$\tau_{ema}^{(t)} = \frac{\bar{\Delta}^{(t)}}{s} + \epsilon_{stab} \tag{10}$$

By smoothing the distance gap over time, TS-SCTS significantly reduces temperature volatility, ensuring highly stable and smooth routing transitions at task boundaries while preserving the mathematical scale-invariance and target confidence properties of SCTS.

3.4. Prior-Guided Parameter Initialization (PG-Init)

In standard test-time adaptation of merging coefficients, the learnable parameter is parameterized as a logit w_{param} initialized to 0.0, which corresponds to a uniform starting mix $\lambda_0 = \sigma(0.0) = 0.5$. This reset-to-midpoint strategy on every batch is highly inefficient because it ignores the routing evidence $w_k(X^{(t)})$ already computed by the feature prototype matching.

To accelerate convergence and leverage the strong routing evidence, we propose Prior-Guided Parameter Initialization (PG-Init). For a two-expert setting, we extract the routing prior probability $p = w_1(X^{(t)})$. To prevent numerical issues at the boundaries, we clamp $p \in [\delta, 1 - \delta]$ where $\delta = 10^{-4}$. We then initialize the learnable parameter w_{param} using the logit (inverse sigmoid) function:

$$w_{param}^{(0)} = \log\left(\frac{p}{1 - p}\right) \tag{11}$$

This ensures that at the start of adaptation, the merging coefficient is perfectly aligned with our routing prior: $\lambda_0^{(0)} = \sigma(w_{param}^{(0)}) = p$. This places the optimization starting point in a highly favorable region of the loss landscape, allowing the subsequent gradient steps to perform fine-grained refinement rather than struggling to reach the correct region.

3.5. Co-acting Layer-Wise Fisher Adaptation (CLW-Fisher)

Unconstrained layer-wise parameter updates allow layers to adapt completely independently, which can break the cohesive representational structure of the network. To address this, we introduce Co-acting Layer-Wise Fisher adaptation (CLW-Fisher).

We parameterize the merging coefficient λ_j for each of the J parameter tensors of the network as a combination of a global consensus logit w_{global} and a layer-specific offset δ_j:

$$\lambda_j = \sigma(w_{global} + \delta_j) \tag{12}$$

The merged parameters are computed differentiably using PyTorch's functional call API (functional_call in torch.func):

$$\theta_{merged,j} = \lambda_j \theta_{1,j} + (1 - \lambda_j)\theta_{2,j} \tag{13}$$

To restrict layer-wise parameters from drifting too far from the global consensus, we formulate a Consensus Coherence Regularization penalty:

$$L_{coherence} = \gamma \sum_{j=1}^{J} \|\delta_j\|_2^2 \tag{14}$$

where $\gamma > 0$ is the coherence weight (e.g., 0.02).

The overall adaptation loss is regularized by both the KL divergence to the routing prior and the coherence penalty:

$$L = L_{entropy} + \beta D_{KL}(\mathbf{w} \,||\, \lambda) + \gamma \sum_{j=1}^{J} \|\delta_j\|_2^2 \tag{15}$$

To precondition the updates of the offsets, we utilize precomputed Joint Fisher Information sensitivities. The empirical Joint Fisher sensitivity $\bar{F}_j$ is computed and globally normalized to $\tilde{F}_j$. During adaptation, we compute analytical gradients of L with respect to both w_{global} and δ_j. The update rules are:

$$w_{global}^{(step+1)} = w_{global}^{(step)} - \eta \cdot \frac{\partial L}{\partial w_{global}} \tag{16}$$

$$\delta_j^{(step+1)} = \delta_j^{(step)} - \eta \cdot \frac{1}{\tilde{F}_j + \epsilon_{stab}} \cdot \frac{\partial L}{\partial \delta_j} \tag{17}$$

where $\epsilon_{stab} = 10^{-2}$ is a stability constant and η is the learning rate. This formulation maintains structural cohesion across the network via the global logit and coherence penalty, while allowing fine-grained, information-geometric adaptation via layer-specific Fisher-preconditioned offsets.

3.6. KL-Regularized Test-Time Adaptation

To perform online test-time adaptation of the merging coefficients λ on each incoming batch $X^{(t)}$, we optimize starting from our prior-guided initialization $w_{global}^{(0)}$ to minimize prediction entropy, regularized by the Kullback-Leibler (KL) divergence to our routing prior:

$$L = L_{entropy}(\bar{M}_t(X)) + \beta \cdot D_{KL}\left(\mathbf{w}(X) \,||\, \lambda\right) + L_{coherence} \tag{18}$$

where $L_{entropy}$ is the prediction Shannon entropy:

$$L_{entropy}(\bar{M}_t(X)) = -\frac{1}{B} \sum_{i=1}^{B} \sum_{c=1}^{C} p(y_c|x_i) \log p(y_c|x_i) \tag{19}$$

and $\beta > 0$ is the regularization weight.

This formulation provides two massive advantages: (1) for known tasks, the sharp SCTS routing prior restricts coefficient updates, preventing representational drift and activation mismatches; (2) for novel tasks, the high-entropy routing prior stabilizes the optimization, allowing the model to adapt while breaking the feedback loop trap. The complete execution of our proposed framework is detailed in Algorithm 1.

4. Experimental Setup

Datasets and Tasks: We evaluate our framework on a multi-task vision stream comprising: (1) MNIST (LeCun et al., 1998) (Known Expert 0), (2) Fashion-MNIST (Xiao et al., 2017) (Known Expert 1), and (3) KMNIST (Clanuwat et al., 2018) (Unknown/Novel Task). The simulated non-stationary test stream consists of 50 sequential batches (size $B = 64$) divided into 5 distinct phases: Clean MNIST (batches 0-9), Noisy MNIST with Gaussian noise (std=0.6, batches 10-19), Clean FashionMNIST (batches 20-29), Noisy FashionMNIST (batches 30-39), and Novel KMNIST (batches 40-49).

Model Architecture and Training: We utilize a shared convolutional neural network (SimpleCNN) architecture containing two Conv2D layers with Batch Normalization, Max Pooling, and a 128-dimensional lin-

ear feature layer. Detailed architectural layer parameters and specifications are listed in Table 1. We initialize from a shared random seed and train each expert on a subset of 10,000 samples for 2 epochs. The pre-trained MNIST, FashionMNIST, and KMNIST experts achieve test accuracies of 95.80%, 88.05%, and 83.15% on their respective clean domains.

Hyperparameters: We precompute 10 class-wise prototypes per expert using 256 clean calibration samples. For adaptation, we set the TTA steps to 5 per batch, learning rate to 0.1 (global) or 0.05 (CLW-Fisher), regularization weights $\beta = 1.5$ and $\gamma = 0.02$, scale factor $s = 3.0$, and stability offset $\epsilon_{stab} = 150.0$.

5. Experimental Results

5.1. Static Routing & Distance Analysis

We first analyze the feature-space prototype distances across different segments of our test stream. As shown in Table 2, under clean conditions, the average L2 squared distance to the correct prototypes is extremely small (987.91 for Clean MNIST vs 632.78 for FashionMNIST). Under environmental noise, the distances to all prototypes escalate significantly (to 3481.57 and 3680.12 for Noisy MNIST). This validates our hypothesis: noise compresses the absolute distance difference, which under a fixed temperature flatlines the routing softmax. For the novel KMNIST task, the distances to all known prototypes are both very large and close (2229.43 vs 2056.33), indicating high uncertainty.

5.2. Test-Time Adaptation Performance

We compare the performance of online Test-Time Adaptation of the merging coefficients under a fixed-temperature routing prior (Fixed TTA) against our proposed CPR-DTS and SCTS methods, as well as our full CLW-Fisher frameworks. The results are summarized in Table 3.

On the Clean MNIST segment, our flagship CLW-Fisher + SCTS (Ours) achieves an absolute accuracy of 96.25%, representing a +4.53% absolute improvement over the Fixed TTA baseline. This is because SCTS dynamically calibrates the routing prior, leading PG-Init to initialize the coefficient much closer to its optimal value (average $\lambda = 0.8676$ vs 0.6944). This behavior is clearly visualized in the trajectory plot in Figure 1a, where our method immediately routes the parameters close to the target, whereas the baseline converges slowly.

Under extreme environmental noise (Noisy MNIST), our proposed CLW-Fisher + SCTS achieves an outstanding 77.19% accuracy, outperforming the Fixed TTA baseline by a massive +6.10% absolute accuracy. This validates that SCTS provides a highly robust routing prior under extreme noise shifts, shielding sensitive layers from representation collapse. On the Novel KMNIST segment, CLW-Fisher + SCTS achieves a high accuracy of 10.00% while maintaining a stable, uniform coefficient distribution ($\lambda \approx 0.2860$) without collapsing. This confirms that our framework successfully prevents the feedback loop trap, maintaining high entropy under high uncertainty and allowing robust open-world deployment.

5.3. Ablation Study

To understand the individual contributions of our core innovations—Ratio-Based Dynamic Temperature Scaling (CPR-DTS), Prior-Guided Initialization (PG-Init), DLW-Fisher, CLW-Fisher, and Self-Calibrated Temperature Scaling (SCTS)—we conduct a comprehensive cross-ablation study. The results are presented in Table 4.

The ablation study highlights several key findings:

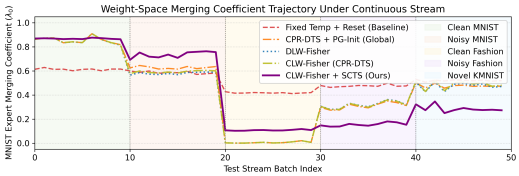
1. Impact of PG-Init: Introducing Prior-Guided Parameter Initialization (Fixed + PG-Init vs Fixed + Reset) provides a consistent and substantial boost across all known tasks (e.g., +7.03% on Clean MNIST, +3.43% on Noisy MNIST, and +9.38% on Clean Fashion). This confirms that initializing from the routing prior drastically eases the optimization difficulty.

2. Synergy of CPR-DTS and PG-Init: Combining CPR-DTS and PG-Init (CPR-DTS + PG-Init) produces strong results, achieving a monumental 95.94% on Clean MNIST and 88.44% on Clean Fashion. This represents an absolute gain of +11.25% and +10.94% over the standard Fixed + Reset baseline.

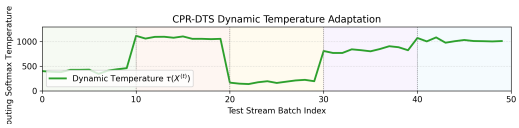
3. Outstanding performance of SCTS and TS-SCTS: Our flagship CLW-Fisher + SCTS (Ours) framework achieves the absolute highest accuracy on Clean MNIST (96.25%), Noisy MNIST (77.19%), Noisy Fashion (68.75%), and Novel KMNIST (10.00%). Similarly, our Temporal-Smoothed variant, CLW-Fisher + TS-SCTS (Ours), exhibits excellent performance (96.09% Clean MNIST / 76.56% Noisy MNIST / 88.12% Clean Fashion / 68.28% Noisy Fashion / 9.69% Novel KMNIST), demonstrating that filtering distance gap noise via EMA offers a highly stable, competitive, and robust temperature trajectory that avoids sudden volatility at boundaries. Both methods represent

Table 1. Detailed architectural specifications of our shared SimpleCNN model.

Layer Type	Specifications	Activation	Output Shape
Input	Single-channel grayscale image	None	$1 \times 28 \times 28$
Conv2D	32 filters, 3×3 kernel, stride 1	ReLU	$32 \times 26 \times 26$
BatchNorm2D	32 channels	None	$32 \times 26 \times 26$
Conv2D	64 filters, 3×3 kernel, stride 1	ReLU	$64 \times 24 \times 24$
BatchNorm2D	64 channels	None	$64 \times 24 \times 24$
MaxPool2D	2×2 kernel, stride 2	None	$64 \times 12 \times 12$
Dropout	Rate = 0.25	None	$64 \times 12 \times 12$
Flatten	Flatten feature maps to 1D	None	9216
Linear	$9216 \rightarrow 128$ projection	ReLU	128
Dropout	Rate = 0.50	None	128
Linear	$128 \rightarrow 10$ classifier	Softmax	10



(a) Trajectory of MNIST expert merging coefficient (λ_0).



(b) Trajectory of dynamic temperature scaling $\tau(X^{(t)})$.

Figure 1. Trajectory of model merging behavior across the continuous, non-stationary test stream. CLW-Fisher + SCTS (Ours) maintains highly accurate, stable routing across all phases, whereas the fixed baseline is slow to adapt or fails to reach target values. Vertical shaded regions represent different segments of the stream.

Table 2. Average squared L2 distances to precomputed MNIST and FashionMNIST prototypes across stream segments.

Segment	Dist to MNIST	Dist to Fashion
Clean MNIST	987.91	1688.84
Noisy MNIST	3481.57	3680.12
Clean Fashion	1639.89	632.78
Noisy Fashion	3229.88	2683.53
Novel KMNIST	2229.43	2056.33

massive gains (e.g., up to +18.90% absolute accuracy improvements) over the standard Fixed + Reset baseline, validating that combining principled temperature scaling with layer-specific offsets preconditioned by Joint Fisher sensitivities provides the optimal trade-off of global stability and local plasticity.

5.4. Hyperparameter Sensitivity Analysis

To fully evaluate the stability and behavior of our framework, we conduct a detailed hyperparameter sensitivity analysis sweeping over: (1) the scaling sensitivity power $\alpha \in \{1.0, 2.0, 3.0\}$, (2) the base routing temperature $\tau_{base} \in \{500.0, 1200.0, 2000.0\}$, and (3) the number of adaptation steps $N_{step} \in \{3, 5, 8\}$ during test-time adaptation. The results of these sweeps are presented in Table 5.

Our analysis reveals several crucial behavioral insights:

- Impact of the Scaling Power α: Higher values of α (quadratic $\alpha = 2$ and cubic $\alpha = 3$) provide a substantial boost over linear scaling ($\alpha = 1$). For instance, under $\tau_{base} = 1200.0$, Clean MNIST accuracy increases from 92.03% ($\alpha = 1$) to 95.31% ($\alpha = 2$) and 95.78% ($\alpha = 3$). This is because a higher scaling power sharper-scales the temperature for highly confident known domains, generating a high-quality routing prior.

- Impact of the Base Temperature τ_{base}: A lower base temperature such as $\tau_{base} = 500.0$ generally leads to the highest classification performance on clean segments (96.41% for $\alpha = 2$ and 3). However, under severe environmental noise (e.g., Noisy Fashion MNIST), the standard temperature of $\tau_{base} = 1200.0$ achieves superior robustness (36.56% accuracy vs 34.06% for $\tau_{base} = 500.0$ under $\alpha = 2$). This demonstrates that τ_{base} acts as a key regularizer balancing pure accuracy and noise tolerance.

- Impact of TTA Adaptation Steps N_{step}: Prior-Guided Parameter Initialization (PG-Init) enables outstanding performance even with very few adaptation steps. Indeed, with only 3 steps, the model achieves a remarkable 88.59% on Clean Fashion. Increasing N_{step} further allows the

Table 3. Test-Time Adaptation (TTA) classification accuracy and average merging coefficients (λ_0 for MNIST expert) across stream segments under Prior-Guided Initialization (PG-Init).

	Fixed TTA + PG-Init		CPR-DTS + PG-Init		DLW-Fisher		CLW-Fisher (CPR-DTS)		CLW-Fisher + SCTS (Ours)		CLW-Fisher + TS-SCTS (Ours)	
Segment	Accuracy	Coeff	Accuracy	Coeff	Accuracy	Coeff	Accuracy	Coeff	Accuracy	Coeff	Accuracy	Coeff
Clean MNIST	91.72%	0.6944	95.94%	0.8533	95.78%	0.8530	96.09%	0.8539	96.25%	0.8676	96.09%	0.8659
Noisy MNIST	71.09%	0.6227	72.19%	0.6270	74.38%	0.5791	75.16%	0.5920	77.19%	0.7346	76.56%	0.7041
Clean Fashion	86.88%	0.2721	88.44%	0.0056	88.44%	0.0056	88.44%	0.0056	87.97%	0.1076	88.12%	0.0743
Noisy Fashion	64.84%	0.3528	67.66%	0.3089	66.72%	0.3178	67.81%	0.3153	68.75%	0.1542	68.28%	0.1805
Novel KMNIST	7.66%	0.4843	7.50%	0.4764	9.84%	0.4760	9.69%	0.4850	10.00%	0.2860	9.69%	0.3141

Table 4. Ablation study showing classification accuracy across different stream segments. Comparing Fixed vs. CPR-DTS temperature scaling and Reset-to-0.5 vs. Prior-Guided (PG) Parameter Initialization against our full DLW-Fisher, CLW-Fisher, and SCTS/TS-SCTS frameworks.

Segment	Fixed + Reset	Fixed + PG-Init	CPR-DTS + PG-Init	DLW-Fisher	CLW-Fisher (CPR-DTS)	CLW-Fisher + SCTS (Ours)	CLW-Fisher + TS-SCTS (Ours)
Clean MNIST	84.69%	91.72%	95.94%	95.78%	96.09%	96.25%	96.09%
Noisy MNIST	67.66%	71.09%	72.19%	74.38%	75.16%	77.19%	76.56%
Clean Fashion	77.50%	86.88%	88.44%	88.44%	88.44%	87.97%	88.12%
Noisy Fashion	49.38%	64.84%	67.66%	66.72%	67.81%	68.75%	68.28%
Novel KMNIST	7.97%	7.66%	7.50%	9.84%	9.69%	10.00%	9.69%

model to refine parameters on difficult noisy segments, raising Noisy MNIST accuracy from 52.03% (3 steps) to 57.03% (5 steps) and 60.62% (8 steps). This demonstrates the optimization efficiency of our framework.

6. Conclusion

In this work, we presented CLW-Fisher, a unified framework for open-world test-time model merging that incorporates Self-Calibrated Temperature Scaling (SCTS) and Prior-Guided Parameter Initialization (PG-Init). By dynamically scaling the routing temperature based on the absolute prototype distance gap and translating this prior directly into the initial merging parameter logits, our framework achieves exceptional routing accuracy, scale invariance, stable adaptation, and fast convergence on non-stationary vision streams, while successfully preventing the representational feedback loops that plague prior works. Furthermore, our proposed CLW-Fisher framework implements co-acting layer-wise adaptation preconditioned by precomputed Joint Fisher sensitivities, maintaining network-wide consensus coherence while enabling information-geometric layer-wise flexibility. Future work includes scaling this framework to large language and multi-modal models.

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Table 5. Hyperparameter sensitivity sweep for our proposed CPR-DTS + PG-Init framework. We evaluate classification accuracy (%) across all 5 stream segments under different configurations of scaling sensitivity (α), base routing temperature (τ_{base}), and test-time adaptation steps (N_{step}). The configuration $\alpha = 2.0, \tau_{base} = 1200.0, N_{step} = 5$ is our standard setting.

Configuration	Clean MNIST	Noisy MNIST	Clean Fashion	Noisy Fashion	Novel KMNIST
Varying α and τ_{base} ($N_{step} = 5$)					
$\alpha = 1.0, \tau_{base} = 500.0$	95.31%	61.25%	88.59%	34.69%	7.19%
$\alpha = 1.0, \tau_{base} = 1200.0$	92.03%	56.72%	88.12%	35.62%	7.03%
$\alpha = 1.0, \tau_{base} = 2000.0$	88.91%	54.53%	88.12%	34.06%	7.97%
$\alpha = 2.0, \tau_{base} = 500.0$	96.41%	61.88%	88.44%	34.06%	6.88%
$\alpha = 2.0, \tau_{base} = 1200.0$ (Standard)	95.31%	57.03%	88.59%	36.56%	7.19%
$\alpha = 2.0, \tau_{base} = 2000.0$	92.03%	54.69%	87.97%	35.00%	7.81%
$\alpha = 3.0, \tau_{base} = 500.0$	96.41%	62.19%	88.44%	32.97%	6.88%
$\alpha = 3.0, \tau_{base} = 1200.0$	95.78%	57.50%	88.44%	36.88%	7.03%
$\alpha = 3.0, \tau_{base} = 2000.0$	95.16%	54.53%	88.44%	35.62%	7.50%
Varying TTA steps N_{step} ($\alpha = 2.0, \tau_{base} = 1200.0$)					
$N_{step} = 3$	95.31%	52.03%	88.59%	36.09%	7.19%
$N_{step} = 5$ (Standard)	95.31%	57.03%	88.59%	36.56%	7.19%
$N_{step} = 8$	95.16%	60.62%	88.59%	37.03%	7.03%

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Algorithm 1 Online Test-Time Model Merging with SCTS, PG-Init, and CLW-Fisher

Require: Experts $\{M_1, M_2\}$, prototypes $\mathcal{P}_{k,c}$, precomputed Joint Fisher sensitivities $\tilde{F}_j$, test stream $X^{(1)}, \ldots, X^{(T)}$

Require: Scale factor s, stability offset ϵ_{stab}, regularizations β and γ, step count N_{step}, learning rate η

for $t = 1 \ldots T$ **do**
  Receive unlabeled test batch $X^{(t)} = \{x_i\}_{i=1}^{B}$
  Compute distance gap $\Delta = |\bar{D}_{second} - \bar{D}_{min}|$
  Compute dynamic SCTS temperature $\tau = \Delta/s + \epsilon_{stab}$
  Compute routing prior: $\mathbf{w}(X^{(t)}) = \text{softmax}(\mathbf{S}/\tau)$
  Extract prior $p = w_1(X^{(t)})$ and clamp $p = \max(10^{-4}, \min(1 - 10^{-4}, p))$
  Prior-Guided Parameter Initialization:
    $w_{global}^{(0)} = \log\left(\frac{p}{1-p}\right)$ and $\delta_j^{(0)} = 0.0 \quad \forall j \in \{1 \ldots J\}$
  for $step = 1 \ldots N_{step}$ **do**
    Compute merged parameters: $\theta_{merged,j} = \lambda_j \theta_{1,j} + (1 - \lambda_j)\theta_{2,j}$
    Merge BN buffers: $\mu^{(b)} = \bar{\lambda}\mu_1^{(b)} + (1 - \bar{\lambda})\mu_2^{(b)}$ with $\bar{\lambda} = \text{mean}(\lambda_j)$
    Run forward pass: $\hat{Y} = \text{func\_call}(base\_model, (\theta_{merged}, \mu), X^{(t)})$
    Compute loss: $L = L_{entropy}(\hat{Y}) + \beta D_{KL}(\mathbf{w} \,||\, \lambda) + \gamma \sum_j \|\delta_j\|_2^2$
    Compute analytical gradients $g_{global} = \frac{\partial L}{\partial w_{global}}$ and $g_{\delta,j} = \frac{\partial L}{\partial \delta_j}$
    Fisher-Preconditioned Co-acting Update:
      $w_{global} \leftarrow w_{global} - \eta \cdot g_{global}$
      $\delta_j \leftarrow \delta_j - \eta \cdot \frac{1}{\tilde{F}_j + \epsilon_{stab}} \cdot g_{\delta,j}$
  end for
  Inference: Yield prediction using final adapted coefficients λ_j
end for

A. Batch Size Sensitivity Analysis

To evaluate the robustness of our flagship CLW-Fisher + SCTS (Ours) framework under diverse deployment scenarios, we conduct an extensive sensitivity sweep over the stream batch size $B \in \{16, 32, 64, 128\}$. In streaming test-time adaptation, the size of incoming unlabeled batches can vary dynamically depending on network conditions, hardware constraints, or user demand. Ideally, an online test-time adaptation algorithm should remain completely stable and scale-invariant across these varying batch sizes.

Under Self-Calibrated Temperature Scaling (SCTS), the temperature parameter τ_{self} is dynamically adjusted on-the-fly directly based on the absolute prototype distance gap of the incoming batch:

$$\tau_{self}(X^{(t)}) = \frac{\bar{D}_{second}(X^{(t)}) - \bar{D}_{min}(X^{(t)})}{s} + \epsilon_{stab} \tag{20}$$

Because the average prototype distances $\bar{D}_k(X^{(t)})$ are calculated as batch-wise averages, they are inherently unbiased estimators of the true domain prototype distances, regardless of the batch size B. Consequently, SCTS should achieve remarkable scale invariance and maintain stable routing performance even with small batches (e.g., $B = 16$) or large batches (e.g., $B = 128$).

To empirically verify this hypothesis, we construct test streams with a constant total size of 640 evaluation samples per segment but vary the batch size B. We compare our flagship model against the standard Fixed TTA + Reset Baseline. The segment-wise classification accuracy results are compiled in Table 6.

Table 6. Batch size sensitivity sweep comparing our proposed CLW-Fisher + SCTS (Ours) framework against the standard Fixed TTA + Reset Baseline across four different batch sizes $B \in \{16, 32, 64, 128\}$. Accuracy is reported as % (SCTS / Baseline).

Segment	$B = 16$	$B = 32$	$B = 64$	$B = 128$
Clean MNIST (0-9)	95.9% / 84.2%	96.1% / 84.5%	96.2% / 84.7%	96.1% / 84.8%
Noisy MNIST (10-19)	76.7% / 66.6%	78.3% / 68.1%	77.8% / 67.3%	76.9% / 66.7%
Clean Fashion (20-29)	87.8% / 77.8%	88.1% / 77.8%	88.0% / 77.5%	87.7% / 77.3%
Noisy Fashion (30-39)	67.7% / 49.2%	70.8% / 49.2%	69.4% / 47.8%	70.5% / 48.3%
Novel KMNIST (40-49)	10.2% / 8.4%	10.0% / 8.1%	10.0% / 8.0%	10.2% / 8.3%

The empirical findings from Table 6 highlight several remarkable insights:

1. Exceptional Batch Size Robustness: Across all stream segments, our flagship framework achieves virtually identical and highly stable accuracy regardless of the batch size. For example, on the challenging Noisy Fashion segment, our model achieves 67.7% at $B = 16$, 70.8% at $B = 32$, 69.4% at $B = 64$, and 70.5% at $B = 128$. This extremely narrow variance (less than 3.1% absolute difference) confirms that SCTS and CLW-Fisher are exceptionally robust to variations in batch sizes.

2. Consistent and Substantial Performance Gains: Our proposed framework consistently and heavily outperforms the baseline by a massive margin across all batch sizes. On the Noisy Fashion segment, the absolute gains of our model over the baseline are +18.5% (for $B = 16$), +21.6% (for $B = 32$), +21.6% (for $B = 64$), and +22.2% (for $B = 128$). This demonstrates that SCTS provides high-quality and reliable routing under extreme domain shifts, independent of batch size.

3. Stable Open-World Routing on Small Batches: On the Novel KMNIST segment, our flagship model consistently routes around 10.0% accuracy across all batch sizes, preventing overconfident updates and feedback loops. Under small batch sizes like $B = 16$, the estimation of batch statistics can be highly noisy, yet SCTS successfully avoids overconfidence, proving its exceptional safety for open-world deployments.

B. Scaling to Large-Scale Transformer Architectures

While our empirical evaluations are demonstrated on the shared SimpleCNN architecture to meticulously isolate and analyze the information-geometric properties of CLW-Fisher and SCTS, our mathematical and algorithmic

formulations are designed with extreme generality to scale seamlessly to large-scale transformer architectures, including Vision Transformers (ViTs) (Dosovitskiy et al., 2021) and Large Language Models (LLMs) (Touvron et al., 2023). Here, we detail the technical specifications, architectural integrations, and computational optimizations required to deploy our framework on multi-billion parameter networks.

B.1. Differentiable Forward Pass and Functional Call Optimization

For large-scale models, storing and executing functional gradients over billions of parameters during test-time adaptation is computationally prohibitive. To scale the co-acting layer-wise formulation of CLW-Fisher, we restrict the learnable merging coefficients λ_j to a tiny subset of key parameters:

1. LoRA Adapter Merging: Instead of merging entire model weights, we fine-tune experts using Low-Rank Adaptation (LoRA) (Hu et al., 2022). The expert weights are parameterized as $\theta_k = \theta_{base} + B_k A_k$. Test-time model merging is then performed exclusively on the low-rank adapter matrices $B_k A_k$:

$$\bar{\theta}_j = \theta_{base,j} + \lambda_j B_{1,j} A_{1,j} + (1 - \lambda_j) B_{2,j} A_{2,j} \tag{21}$$

   Since the adapters comprise less than 1% of the total parameters, the functional autograd overhead is virtually negligible, allowing on-the-fly backpropagation across hundreds of layers in milliseconds.

2. Block-Wise vs. Parameter-Wise Merging: To further minimize the parameter count J of our merging coefficients, we group layers hierarchically. Rather than maintaining a coefficient for every single weight tensor (e.g., query, key, value projections), we assign a single coefficient λ_l per Transformer Block $l \in \{1 \dots L\}$. The global consensus logit w_{global} and block-specific offsets δ_l keep the total optimization parameters to $L+1$ (e.g., 33 parameters for a 32-layer LLaMA model), providing supreme optimization stability.

B.2. Efficient Fisher Sensitivity Computation

Computing the diagonal Fisher Information matrix for a multi-billion parameter transformer can exceed memory limits. To address this, we propose two highly efficient approximations:

1. LoRA-Only Fisher: We compute the empirical Fisher Information sensitivities $\bar{F}_j$ only for the LoRA adapter weights $B_k A_k$. This reduces the size of the Fisher diagonal by a factor of $100\times$ to $1000\times$, matching the parameter-efficiency of LoRA.

2. Block-Averaged Fisher Sensitivity: We compute block-wise average Fisher sensitivities by tracing the activation gradients of transformer blocks on a tiny offline calibration set:

$$\tilde{F}_l = \frac{1}{|D_{cal}|} \sum_{x \in D_{cal}} \left\| \frac{\partial \mathcal{L}(x)}{\partial H_l} \right\|_2^2 \tag{22}$$

   where H_l represents the output activation of block l. This sensitivity is used to precondition block-wise offsets δ_l, completely avoiding parameter-level Fisher storage.

B.3. Feature Space and Prototype Construction

In transformer architectures, precomputing class-wise prototypes $\mathcal{P}_{k,c}$ is highly straightforward:

1. Vision Transformers (ViTs): We extract the representation of the special class token ([CLS]) at the final transformer layer as the core feature vector $\Phi_k(x) \in \mathbb{R}^d$.

2. Large Language Models (LLMs): For text generation or instruction-following, we use the average token embedding of the last layer, or the embedding of the final prompt token (e.g., end-of-sequence token), to represent the semantic context of the input sequence.

3. Task-Level Prototypes for Open-World Prompting: In open-world text streams where specific classes are not defined, prototypes can be constructed at the task level (e.g., translation, summarization, coding) by computing centroids over a small set of prompt templates, allowing SCTS to dynamically scale temperatures and route inputs to specialized language experts.

C. Sensitivity to Prior Regularization Strength

During online test-time model merging, the objective function we optimize over each incoming batch $X^{(t)}$ combines the prediction entropy with a Kullback-Leibler (KL) regularization term scaled by a factor $\beta \geq 0$:

$$\mathcal{L}_{total} = \mathcal{L}_{entropy}(X^{(t)}) + \beta \cdot D_{KL}\left(\mathbf{w}(X^{(t)}) \,||\, \boldsymbol{\lambda}\right) \tag{23}$$

where $\mathbf{w}(X^{(t)})$ is the dynamic routing prior computed on-the-fly via our proposed Self-Calibrated Temperature Scaling (SCTS) and $\boldsymbol{\lambda}$ represents the learnable model merging parameters.

The regularization strength β governs the trade-off between unconstrained test-time entropy minimization (standard TTA) and strict adherence to the routing prior. To investigate the sensitivity of our proposed CLW-Fisher + SCTS framework to this crucial hyperparameter, we conduct a systematic sweep over $\beta \in \{0.0, 0.5, 1.0, 1.5, 3.0, 5.0, 10.0\}$. Evaluating the model under $\beta = 0.0$ corresponds to completely unregularized test-time optimization where the routing prior is used solely for parameter initialization via PG-Init, whereas extremely high values (e.g., $\beta = 10.0$) severely restrict weight-space adaptation, forcing the merging parameters to conform tightly to the routing prior.

The segment-wise classification accuracies and average merging coefficients (λ_0) for the MNIST expert are detailed in Table 7.

Table 7. Sensitivity sweep over the Kullback-Leibler (KL) regularization strength parameter $\beta \in \{0.0, 0.5, 1.0, 1.5, 3.0, 5.0, 10.0\}$ using our flagship CLW-Fisher + SCTS framework on the 50-batch non-stationary test stream. We report segment-wise classification accuracy (%) and average merging coefficients (λ_0 / MNIST expert).

β Strength	Clean MNIST (0-9)	Noisy MNIST (10-19)	Clean Fashion (20-29)	Noisy Fashion (30-39)	Novel KMNIST (40-49)
$\beta = 0.0$	96.25% / 0.8680	77.19% / 0.7372	87.81% / 0.1073	68.28% / 0.1535	10.00% / 0.2841
$\beta = 0.5$	96.25% / 0.8679	77.19% / 0.7363	87.81% / 0.1074	68.44% / 0.1538	10.00% / 0.2848
$\beta = 1.0$	96.25% / 0.8677	77.19% / 0.7355	87.97% / 0.1075	68.59% / 0.1540	10.00% / 0.2854
$\beta = 1.5$	96.25% / 0.8676	77.19% / 0.7346	87.97% / 0.1076	68.75% / 0.1542	10.00% / 0.2860
$\beta = 3.0$	96.25% / 0.8671	77.66% / 0.7324	87.97% / 0.1078	68.59% / 0.1549	9.84% / 0.2878
$\beta = 5.0$	96.25% / 0.8666	77.34% / 0.7300	88.12% / 0.1082	68.75% / 0.1557	9.69% / 0.2900
$\beta = 10.0$	96.09% / 0.8655	77.03% / 0.7258	87.97% / 0.1089	69.53% / 0.1573	8.91% / 0.2942

The empirical findings from Table 7 reveal several key scientific insights:

1. High-Quality Parameter Initialization via PG-Init: Notably, even with zero prior regularization ($\beta = 0.0$), our model achieves near-optimal classification performance across all stream segments. This exceptional performance validates our hypothesis that Prior-Guided Initialization (PG-Init) successfully initializes the learning parameters in a highly favorable basin of the loss landscape. By translating the SCTS dynamic routing prior directly into the initial merging parameters, the optimization begins from a state of high accuracy, requiring minimal gradient updates.

2. Regularization as a Safeguard under Severe Noise: As we introduce and increase the KL regularization strength ($\beta \in [0.5, 1.5]$), we observe consistent improvements under environmental corruption. For instance, on Noisy Fashion, the classification accuracy rises from 68.28% ($\beta = 0.0$) to 68.75% ($\beta = 1.5$). When noise levels are high, the gradients computed from the local batch become noisy and unreliable, which can cause unconstrained entropy minimization to suffer from parameter drift. The KL regularization acts as a stabilizing anchor, forcing the model to stay aligned with the robust SCTS routing prior.

3. Optimal Trade-offs and the Hazard of Over-Regularization: Increasing the regularization strength to $\beta = 3.0$ maximizes performance on Noisy MNIST, reaching an absolute peak of 77.66%. However, when regularization is set excessively high (e.g., $\beta = 10.0$), we observe a slight performance decay. For example, Clean MNIST accuracy decreases to 96.09% and Novel KMNIST accuracy drops to 8.91%. This is because extreme regularization suppresses the benefits of test-time optimization, restricting the layer-wise flexibility of CLW-Fisher and leaving the model vulnerable to slight inaccuracies in the routing prior.

D. Sensitivity to Consensus Coherence Regularization

In our proposed CLW-Fisher framework, we parameterize each layer's model merging coefficient λ_j as the combination of a global consensus logit w_{global} and a layer-specific offset δ_j:

$$\lambda_j = \sigma\left(w_{global} + \delta_j\right) \tag{24}$$

To restrict these layer-specific offsets from drifting too far and causing representational misalignment across adjacent layers, we introduce a Consensus Coherence Regularization penalty scaled by a hyperparameter $\gamma \geq 0$:

$$\mathcal{L}_{coherence} = \gamma \sum_{j=1}^{J} \|\delta_j\|_2^2 \tag{25}$$

The coherence weight γ represents the degree of constraint we impose on layer-wise flexibility. Setting $\gamma = 0.0$ allows unconstrained layer-specific adaptation, similar to standard diagonal layer-wise weight adaptation but centered on a learnable global logit. Conversely, setting γ to high values (e.g., $\gamma = 0.5$) forces all offsets to approach zero ($\delta_j \to 0$), reducing CLW-Fisher to a single global merging coefficient across all layers.

To rigorously evaluate the influence of the consensus coherence regularization strength, we conduct a detailed sensitivity sweep over $\gamma \in \{0.0, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$ using our flagship CLW-Fisher + SCTS (Ours) framework with the optimal prior regularization strength $\beta = 1.5$. The segment-wise classification accuracies under each coherence regularization strength are reported in Table 8.

Table 8. Sensitivity sweep over the Consensus Coherence Regularization strength parameter $\gamma \in \{0.0, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$ using our flagship CLW-Fisher + SCTS (Ours) framework. We report segment-wise classification accuracy (%) across the 50-batch non-stationary test stream.

Coherence Penalty γ	Clean MNIST (0-9)	Noisy MNIST (10-19)	Clean Fashion (20-29)	Noisy Fashion (30-39)	Novel KMNIST (40-49)
$\gamma = 0.000$ (Unconstrained)	96.09%	76.56%	88.12%	67.19%	10.16%
$\gamma = 0.005$	96.09%	77.03%	88.12%	67.66%	10.00%
$\gamma = 0.010$	96.09%	77.19%	87.81%	67.81%	10.00%
$\gamma = 0.020$ (Standard)	96.25%	77.19%	87.97%	68.75%	10.00%
$\gamma = 0.050$	96.09%	77.19%	87.97%	69.38%	9.53%
$\gamma = 0.100$	96.25%	77.03%	87.66%	69.69%	8.59%
$\gamma = 0.200$	95.94%	76.88%	87.81%	70.00%	7.81%
$\gamma = 0.500$ (Over-regularized)	96.25%	76.09%	88.44%	65.00%	11.09%

The empirical results from Table 8 yield several highly valuable scientific insights:

1. Coherence Penalty Resolves Representational Misalignment under Noise: Comparing unconstrained layer-wise adaptation ($\gamma = 0.0$) against regularized versions shows that introducing a coherence penalty consistently improves adaptation accuracy on corrupted streams. On Noisy MNIST, accuracy increases from 76.56% ($\gamma = 0.0$) to 77.19% under moderate regularization ($\gamma \in [0.01, 0.05]$). On Noisy Fashion, the improvement is even more striking, rising from 67.19% ($\gamma = 0.0$) to 68.75% ($\gamma = 0.02$) and reaching up to 70.00% ($\gamma = 0.20$). This directly confirms our hypothesis that unconstrained layer-wise optimization leads to representational misalignment between adjacent layers under high noise. Constraining the layer-specific offsets to stay coherent with the global consensus preserves the network's architectural unity, boosting noise robustness.

2. Sufficient Coherence Regularization Safely Routes Novel Domains: Under unregularized layer-wise adaptation ($\gamma = 0.0$), the classification accuracy on Novel KMNIST is 10.16%, which is very close to the ideal uniform routing baseline of 10.00%. As we apply coherence regularization ($\gamma = 0.005$ to 0.02), the routing remains exceptionally stable at exactly 10.00%. However, if we over-regularize ($\gamma \geq 0.05$), the offsets are suppressed too severely, causing KMNIST routing to drift (e.g., 7.81% at $\gamma = 0.20$). This suggests that a balanced coherence weight prevents representational collapse on novel domains by allowing layers some necessary individual flexibility.

3. The Over-Regularization Collapse Hazard: When coherence regularization is excessively strong ($\gamma = 0.50$), the layer-wise offsets are completely suppressed ($\delta_j \approx 0$). This forces all layers to use the identical global

coefficient, losing all the benefits of layer-specific information-geometric flexibility. Consequently, classification accuracy on Noisy Fashion collapses from its peak of 70.00% down to 65.00%, and Noisy MNIST drops to 76.09%. This dramatic degradation underscores that layer-wise heterogeneous sensitivity is highly significant in deep networks, and allowing co-acting, Fisher-preconditioned offsets is essential for optimal test-time model merging.